

NIRJOR SAHA RUDRO

AI Engineer | Computer Vision & Deep Learning

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Dhaka, Bangladesh



PROFILE

AI Engineer at Deep Mind Labs with strong experience in computer vision, deep learning, and LLM-based systems. Previously worked as a Research Assistant in the Deep Research Group at the University of Rajshahi. Passionate about applying cutting-edge AI technologies to solve real-world and research-driven challenges.

EXPERIENCE

AI Engineer

Deep Mind Labs

Apr 2025 – Present Dhaka, Bangladesh

- Integrated Large Language Models using LangChain and LangGraph in research and healthcare domains
- Built scalable orchestration pipelines using MCP
- Developed production-grade Retrieval-Augmented Generation (RAG) systems
- Designed and deployed real-time computer vision pipelines for detecting and classifying items on moving roller conveyors
- Developed backend REST APIs and real-time WebSocket services using Django and FastAPI
- Containerized and deployed AI systems using Docker for scalable and reproducible environments
- Collaborated cross-functionally to deliver end-to-end AI solutions

AI Intern

Deep Mind Labs

Feb 2025 – Mar 2025 Dhaka, Bangladesh

- Applied LangChain and LangGraph for AI-powered applications
- Fine-tuned Large Language Models for task-specific optimization
- Designed automated AI workflows using n8n
- Assisted in deployment and evaluation of AI services

PROJECTS

Multilingual RAG (Bangla & English)

Aug 2025

- Developed a bilingual RAG pipeline supporting Bangla and English queries
- Implemented hybrid retrieval using FAISS and BM25
- Deployed APIs using FastAPI

Python LangChain FAISS FastAPI

Transformer-based Segmenter for Small Datasets

BSc Thesis

Feb 2023 – Nov 2024

- Designed data-efficient Transformer models for retinal blood vessel segmentation
- Built preprocessing pipelines for medical image datasets

EDUCATION

BSc in Computer Science & Engineering

University of Rajshahi

2019 – 2024

CGPA: 3.22 / 4.00

Higher Secondary Certificate (HSC)

Shaparan Govt. College

2016 – 2018

GPA: 4.69 / 5.00

Secondary School Certificate (SSC)

Ram Sundar Pilot High School

2014 – 2016

GPA: 4.72 / 5.00

SKILLS

Python Django FastAPI Docker
TensorFlow PyTorch Transformers
CNN RNN LSTM ViT
LangChain LangGraph RAG
Computer Vision SQL Linux Git
Unity AR

CERTIFICATIONS

- Git & GitHub for Open Source Collaboration
Coursera
- Introduction to C# Programming and Unity
Coursera
- Build Your First Android App
Coursera
- Programming Foundations with JavaScript, HTML, and CSS
Coursera
- Introduction to Game Development
Coursera

TensorFlow

Transformers

Medical Imaging

Wav2Lip Implementation (TensorFlow)

📅 Nov 2022 – Jan 2023

- Reimplemented Wav2Lip from PyTorch to TensorFlow
- Addressed framework-specific weight initialization and layer behavior issues

TensorFlow

PyTorch

Game Development Projects (Unity)

Tower Defense, 2D Games, AR

📅 2021 – 2022

- Developed a strategy-based tower defense game with dynamic enemy waves and upgrades
- Built additional 2D platformer and AR-based gameplay experiences

Unity

C#

AR

PUBLICATION

Competency Comparison of Deep Neural Networks for Identifying Gender in Color Fundus Photographs

ICCIT 2023

📅 Nov 2023

📍 Cox's Bazar, Bangladesh

- Published at the 26th International Conference on Computer and Information Technology
- DOI: 10.1109/ICCIT60459.2023.10441623